Haxe: Write Once, Convert to Many Languages

The proliferation of a wide variety of platforms has become a major challenge for developers when it comes to porting their application to different environments. Haxe assists developers to face this challenge. This article is an eye opener to the world of cross-compilation based programming using the Haxe toolkit.

One of the major challenges for software developers today is dealing with the variety of software environments or platforms. Applications can enjoy the fullest potential of the platforms only when they are built using native codebase. To harness this advantage, organisations need to have developers who are skilled in those individual development techniques. Typically, teams address this with the help of coders fluent in specific programming languages to port an application to a specific platform. Now, instead of this traditional approach, there are special multi-platform or multi-paradigm cross-compiler based languages.

Multi-platform languages

Here, the word 'multi-platform' is not used to imply that the language will work on different platforms, but rather that the source code of these languages can be compiled into the source code of other programming languages. This can also be called 'source-to-source compilation'. Two such popular languages are Haxe and Monkey. The code written in these programming languages is compiled into specific target languages, which can again be handled as usual programs written in those target programming languages. Due to this source-to-source translation methodology, all that a developer needs to do is to write the Haxe code and convert it into the target language, so that the platform-specific features can be harnessed fully.

The Haxe toolkit

Haxe is a popular multi-platform toolkit that has five major components:

- The Haxe programming language
- A cross-compiler
- The Haxe library
- Haxe based frameworks
- Additional tools

Haxe is a high-level programming language. With respect to data types, Haxe is strictly a typed language. The effort required to learn the syntax of Haxe is very minimal if you already know any of the modern day programming languages like Java, PHP, etc.

Why Haxe?

Though there are other alternatives to Haxe in the multi-platform programming world, Haxe possesses the following advantages: